

THE THEORY OF ENTROPICITY (ToE)

LIVING REVIEW LETTERS SERIES

SUPPLEMENTARY APPENDIX K.1 TO LETTER IE

Appendix K.1: The Schwarzschild Solution from the Generalized Entropic Field Equations / Obidi Field Equations (GEFE/OFE)

*The Unique Static, Spherically Symmetric Vacuum Geometry as an Emergent
Classical Limit of the Theory of Entropicity*

John Onimisi Obidi

Research Lab, The Aether

jonimisiobidi@gmail.com

May 2026

Abstract

This appendix presents a complete, self-contained derivation of the Schwarzschild solution — the unique static, spherically symmetric vacuum geometry of general relativity — from the Generalized Entropic Field Equations (GEFE), equivalently designated the Obidi Field Equations (OFE), as formulated within the Theory of

Entropicity (ToE). The derivation proceeds in four logically sequential steps: (1) solving the Minimal Entropic Equation (MEE) for a static, spherically symmetric entropic field configuration; (2) constructing the Fisher-Rao information metric on the statistical manifold induced by the entropic field; (3) applying the emergence map to obtain the physical spacetime metric; and (4) identifying the Schwarzschild mass parameter as an emergent entropic charge. The result demonstrates that the Schwarzschild geometry is not an independent postulate within the ToE framework, but rather an *emergent* classical consequence of an underlying entropic field configuration on the information manifold. The mass of a gravitating body is thereby reinterpreted as a measure of the entropic charge concentrated in its field profile. This appendix serves as the first concrete worked example of the GEFE/OFE producing a known exact solution of Einstein's general relativity, and constitutes a foundational validation of the emergence mechanism articulated in Letter IE of the ToE Living Review Letters Series.

1. Introduction

1.1. The Schwarzschild Solution in General Relativity: Historical and Physical Context

In the annals of theoretical physics, few results carry the foundational weight of the Schwarzschild solution. Barely two months after Albert Einstein published his field equations of general relativity in November 1915 — the

culminating achievement of a decade of intensive intellectual struggle — Karl Schwarzschild, writing from the Eastern Front of the First World War, produced the first exact, closed-form solution to those equations. His paper, communicated to the Prussian Academy of Sciences on 13 January 1916, described the gravitational field outside a static, spherically symmetric, non-rotating, uncharged mass distribution. The solution was communicated by Einstein himself, who reportedly expressed astonishment that an exact solution could be found at all, given the formidable nonlinearity of the field equations. Tragically, Schwarzschild died only months later, in May 1916, of the autoimmune disease pemphigus, contracted during his military service. His solution, however, endures as one of the most consequential results in all of mathematical physics.

The Schwarzschild metric, expressed in the coordinates that bear his name, takes the form

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

rc

2

))

c

2

dt

2

+ (1

-

2

GM

/(

rc

2

))

-1

dr

2

+

r

2

(

$$d\theta^2 + \sin^2\theta d\varphi^2)$$

where G is Newton's gravitational constant, M is the mass of the gravitating body, c is the speed of light, and (t, r, θ, φ) are the Schwarzschild coordinates. This metric describes the unique spacetime geometry outside any static, spherically symmetric, non-rotating, electrically neutral mass distribution. Its domain of applicability is breathtaking: it governs the gravitational field around ordinary stars such as the Sun, around planets, around neutron stars, and — in its maximally extended form — it describes the exterior and interior of non-rotating (Schwarzschild) black holes. Every measurement of gravitational phenomena in the solar system, from the precession of Mercury's perihelion to the Shapiro time delay of radar signals grazing the Sun, from the deflection of starlight by the solar gravitational field (first confirmed by Eddington's 1919 eclipse expedition) to the gravitational redshift of photons climbing out of a gravitational potential well (confirmed by the Pound-Rebka experiment of 1960), constitutes a test of the Schwarzschild solution.

The Schwarzschild metric possesses two distinguished radial scales. The first is the Schwarzschild radius, $r_s = 2GM/c^2$, at which the temporal metric coefficient vanishes and the radial metric coefficient diverges — a coordinate singularity that marks the event horizon of a black hole. The second is the genuine curvature singularity at $r = 0$, where the Kretschmer scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/(c^4r^6)$ diverges, signaling a breakdown of the classical description. For ordinary astrophysical bodies (stars, planets), the physical surface lies well outside the Schwarzschild radius, and only the exterior solution is physically relevant. For black holes, the full Schwarzschild geometry — including the event horizon and the interior — becomes essential.

The centrality of the Schwarzschild solution to gravitational physics cannot be overstated. It is the zeroth-order geometry around which all perturbative calculations in black hole physics, gravitational wave theory, and post-Newtonian celestial mechanics are organized. The quasi-normal modes of a Schwarzschild black hole, the Regge-Wheeler and Zerilli equations governing its perturbations, the Hawking temperature $T_H = \hbar c^3/(8\pi k_B GM)$, and the Bekenstein-Hawking entropy $S_{BH} = k_B c^3 A/(4G\hbar)$ — all are built upon the Schwarzschild foundation. Any theory of gravity that aspires to physical relevance must, as a minimal requirement, recover the Schwarzschild solution in the appropriate limit.

1.2. Birkhoff's Theorem and the Uniqueness of the Schwarzschild Geometry

In 1923, George David Birkhoff proved a theorem of extraordinary power and economy: the Schwarzschild solution is the *unique* spherically symmetric vacuum solution of Einstein's field equations. This theorem — Birkhoff's theorem — has no analogue in Newtonian gravity. In Newtonian theory, the

gravitational potential outside a spherically symmetric mass distribution is $\Phi = -GM/r$ by virtue of Newton's shell theorem (which Gauss's law for gravity guarantees), but the Newtonian theory does not constrain the time-dependence of the exterior field in the way that Birkhoff's theorem does.

The content of Birkhoff's theorem is more subtle than is often appreciated. It states not merely that the exterior field of a static, spherically symmetric body is given by the Schwarzschild metric — this much might be expected on grounds of symmetry — but that *any* spherically symmetric vacuum region of spacetime, whether or not the matter distribution is static, is described by the Schwarzschild geometry. A spherically symmetric star undergoing purely radial pulsations, for instance, produces no gravitational waves and generates an exterior field that is exactly Schwarzschild at every instant. This is because the only spherically symmetric gravitational degree of freedom — the mass monopole — is conserved by the Bianchi identity, and hence cannot radiate. Birkhoff's theorem is thus a rigidity result: spherical symmetry combined with the vacuum Einstein equations is so constraining that only a one-parameter family of solutions (parameterized by the mass M) exists.

The significance of Birkhoff's theorem for the present appendix is twofold. First, it guarantees that if the GEFE/OFE reduce to the vacuum Einstein equations in the appropriate classical limit, then the resulting geometry is *necessarily* Schwarzschild — the uniqueness is automatic. Second, it implies that the entropic field configuration that generates the Schwarzschild geometry must itself be unique: the only static, spherically symmetric solution of the Minimal Entropic Equation in the massless, minimal-coupling limit is the Coulomb-like profile $S(r) = S_0 + S_I/r$. This is the entropic analogue of Birkhoff's theorem, and we shall comment on it further in the Remarks at the end of this appendix.

1.3. Recovery of the Schwarzschild Solution as a Critical Test of the GEFE/OFE

Any proposed theory of gravitation — whether it supplements, modifies, or replaces Einstein's general relativity — must confront the Schwarzschild solution as its first and most demanding consistency test. The reason is empirical: the Schwarzschild metric has been confirmed to extraordinary precision by solar-system tests (the post-Newtonian parameter γ has been measured to agree with the GR prediction $\gamma = 1$ to one part in 10^5 by the Cassini spacecraft), by binary pulsar observations (the Hulse-Taylor pulsar PSR B1913+16), and most recently by direct imaging of the photon ring around the supermassive black hole M87* by the Event Horizon Telescope. A theory that cannot recover Schwarzschild in the weak-field, slow-motion, classical limit is empirically falsified before it begins.

The Generalized Entropic Field Equations (GEFE), equivalently the Obidi Field Equations (OFE), constitute the gravitational sector of the Theory of Entropicity. As formulated in Letter IE of the ToE Living Review Letters Series, the GEFE contain the standard Einstein tensor on the left-hand side and, on the right-hand side, a hierarchy of entropic contributions: the Leading-Order Approximation (LOA) terms encoding the kinetic energy, potential energy, and non-minimal coupling of the entropic field; the Second-Order Approximation (SOA) terms encoding higher-curvature corrections derived from the Seeley-DeWitt heat kernel expansion; and the Obidi Curvature Invariant (OCI) tensor, encoding the mismatch between the information manifold curvature and the physical spacetime curvature. The full GEFE are considerably richer than the Einstein equations, and in general they predict corrections to GR at short distances, high curvatures, and quantum scales. However, in the classical, large-scale, weak-field limit — the regime probed by solar-system tests and by the exterior geometry of ordinary

astrophysical bodies — the GEFE must and do reduce to the vacuum Einstein equations. It is in this limit that we shall recover the Schwarzschild solution.

1.4. Strategy of this Appendix

The strategy of this appendix is carefully circumscribed. We do *not* re-derive every step of general relativity from first principles — the Einstein field equations, the Bianchi identity, the post-Newtonian formalism, and so forth are taken as established results of the classical limit of the GEFE. What we *do* demonstrate is that a static, spherically symmetric entropic field configuration, processed through the Fisher-Rao information metric and the emergence map $\Phi: M_I \rightarrow M_S$, yields the Schwarzschild geometry in the classical limit, with the mass parameter M identified as an emergent entropic charge. The derivation proceeds in four steps:

1. **Step 1 (Radial Entropic Field Equation):** We solve the Minimal Entropic Equation (MEE) for a static, spherically symmetric entropic field $S = S(r)$ in the weak-field, minimal-coupling regime, obtaining the Yukawa form and its massless (Coulomb-like) limit.
2. **Step 2 (Fisher-Rao Information Metric):** We construct the Fisher-Rao metric on the statistical manifold induced by the entropic field $S(r)$, using a Gaussian statistical model, and derive the radial and angular components of the information metric.
3. **Step 3 (Emergence Map and Physical Metric):** We apply the emergence map to identify the physical spacetime metric with the information metric, and invoke the classical limit of the GEFE (which reduces to the vacuum Einstein equations) to determine the full Schwarzschild geometry.

4. **Step 4 (Mass Identification):** We match the entropic field parameters to the Schwarzschild mass parameter, identifying M as an emergent entropic charge proportional to the integration constant S_I .

1.5. Connection to Letter IE and the Curvature Transfer Theorem

This appendix is a concrete worked example of the central theorem of Letter IE: the *Curvature Transfer Theorem* (Theorem 3.1). The Curvature Transfer Theorem states that, under the emergence map Φ , the Riemann curvature tensor of the physical spacetime (M_S, g_S) is determined by the pushforward of the Riemann curvature tensor of the information manifold (M_I, g_I) , up to corrections of order $O(1/N)$ where N is the number of entropic degrees of freedom. In the classical limit ($N \rightarrow \infty$), the curvature transfer is exact: the physical spacetime curvature is the information curvature, rescaled by the emergence constant λ .

The Schwarzschild derivation that follows is, in this light, a verification of the Curvature Transfer Theorem in the simplest non-trivial case: a static, spherically symmetric entropic configuration. The information manifold is curved (with curvature determined by the Fisher-Rao metric of the entropic profile $S(r) = S_0 + S_I/r$), and this curvature is transferred, via the emergence map, to the physical spacetime, yielding the Schwarzschild geometry. The Schwarzschild curvature singularity at $r = 0$ is thereby reinterpreted as a singularity of the Fisher-Rao metric — a point at which the information content of the entropic field diverges — and the event horizon at $r = r_s$ is reinterpreted as an entropic surface of distinguished informational significance.

The derivation also connects to other elements of the ToE framework established in the Living Review Letters. The identification of mass as an entropic charge (Step 4) is analogous to the identification of electric charge as a topological winding number of the entropic phase field, established in Letter IIA. The Alemoh-Obidi Correspondence, articulated in Letter IC, provides the philosophical scaffolding: physical quantities are *not* fundamental but emergent from informational and entropic substrates. The present appendix provides the most direct calculational illustration of this correspondence to date.

2. Assumptions

Before proceeding to the derivation, we state and motivate three physically transparent assumptions that define the regime in which the Schwarzschild solution is recovered from the GEFE/OFE. Each assumption corresponds to a well-defined limit of the full ToE formalism, and each is necessary for the reduction of the GEFE to the vacuum Einstein equations.

Assumption 1: Static, Spherically Symmetric Entropic Field

The entropic field depends only on the radial coordinate: $S = S(r)$.

The first assumption restricts the entropic field to the simplest spatial configuration compatible with an isolated, non-rotating gravitating body. The assumption of spherical symmetry — invariance under the full rotation group $SO(3)$ — means that the entropic density is distributed isotropically around a central concentration, with no preferred angular direction. The assumption of

staticity — independence of the time coordinate — means that the entropic field has reached a state of thermodynamic equilibrium: there are no entropic currents, no time-dependent fluctuations, and no dissipation. Taken together, these assumptions reduce the entropic field from a function of four spacetime coordinates, $S(t, r, \theta, \varphi)$, to a function of a single variable, $S(r)$.

This restriction is the entropic analogue of the standard symmetry assumptions used in deriving the Schwarzschild solution in general relativity. Just as the most general static, spherically symmetric metric can be written in terms of two functions of r alone (the temporal and radial metric coefficients $A(r)$ and $B(r)$), the most general static, spherically symmetric entropic field is a function $S(r)$ alone. The reduction from a partial differential equation (the full MEE in four dimensions) to an ordinary differential equation (the radial MEE) is a direct consequence of this symmetry.

Physically, Assumption 1 models the gravitational field of an isolated, non-rotating, uncharged body — a star, a planet, or a non-rotating black hole — viewed from the exterior, at distances large compared to the body's radius. In the ToE framework, this configuration corresponds to a single, isolated concentration of entropic charge surrounded by an entropic vacuum. The entropic field $S(r)$ encodes the radial distribution of information: it is large near the central body (where the microscopic degrees of freedom are densely concentrated) and decays toward the asymptotic entropic background S_0 at spatial infinity.

Assumption 2: Minimal Coupling Regime

The non-minimal coupling parameter vanishes: $\xi \rightarrow 0$.

The second assumption concerns the coupling between the entropic field and the spacetime curvature. In the full Obidi Action (the action functional from which the GEFE are derived), the entropic field couples to the Ricci scalar R through a non-minimal coupling term of the form ξRS^2 . The dimensionless coupling constant ξ parameterizes the strength of this interaction. When $\xi = 0$, the entropic field is *minimally coupled* to gravity: it interacts with the spacetime geometry only through the covariant derivative (i.e., through the metric in the d'Alembertian operator) and through the effective stress-energy tensor that it contributes to the right-hand side of the GEFE.

The minimal coupling regime $\xi \rightarrow 0$ is the simplest and most well-studied case. It is the regime in which the entropic field behaves as a standard minimally coupled scalar field, analogous to the scalar fields studied extensively in scalar-tensor theories of gravity (Brans-Dicke theory, Horndeski theory, etc.) in the limit where the scalar-curvature coupling is turned off. The special value $\xi = 1/6$ corresponds to *conformal coupling* in four dimensions, at which the scalar field equation is conformally invariant on a conformally flat background. While conformal coupling has mathematical elegance and physical interest (it preserves the conformal structure of the equations of motion), the Schwarzschild solution is most straightforwardly recovered in the minimal coupling limit. Non-minimal coupling ($\xi \neq 0$) would produce corrections to the Schwarzschild geometry — scalar hair, modified horizons, and deviations in the post-Newtonian parameters — which constitute genuine predictions of the Theory of Entropicity beyond standard general relativity. These corrections, and their observational consequences, are deferred to future work (see Remark K.1.1).

It is important to emphasize that Assumption 2 is an approximation, not a dogma. The full ToE formalism accommodates arbitrary values of ξ , and indeed the renormalization group flow of ξ under quantum corrections is a

significant open question. The minimal coupling limit is adopted here because our goal is to recover the *standard* Schwarzschild solution, and any non-zero ξ would modify the result. The consistency of this approximation is confirmed *a posteriori* by the fact that the resulting geometry is Ricci-flat ($R = 0$), so that the non-minimal coupling term ξRS^2 vanishes identically — the assumption is self-consistent.

Assumption 3: Classical, Weak-Field, Large-Scale Limit

The GEFE reduce to the Einstein field equations in vacuum.

The third assumption specifies the regime in which the full complexity of the GEFE is suppressed, leaving only the vacuum Einstein equations. This is the classical limit of the Theory of Entropicity, and it is defined by a hierarchy of conditions. First, the *Second-Order Approximation* (SOA) corrections — higher-curvature terms arising from the Seeley-DeWitt heat kernel expansion of the one-loop effective action — are suppressed by powers of $(l_P/L)^2$, where $l_P = (\hbar G/c^3)^{1/2} \approx 1.6 \times 10^{-35}$ m is the Planck length and L is the characteristic macroscopic scale of the problem. For any astrophysical application — solar-system scales, stellar scales, or even black hole event horizons of stellar mass or larger — we have $l_P/L \ll 1$, and the SOA corrections are utterly negligible.

Second, the *Obidi Curvature Invariant* (OCI) tensor vanishes in the classical limit. The OCI measures the curvature mismatch between the information manifold (M_I, g_I) and the physical spacetime (M_S, g_S) . When the emergence map is an isometry (or a conformal isometry with constant conformal factor), the curvature of g_S is fully determined by the curvature of g_I , and the OCI vanishes. This is precisely the situation in the classical regime, where quantum fluctuations do not perturb the emergence map. In the quantum

regime, the OCI becomes non-zero and contributes additional curvature — the "dark curvature" discussed in Letter IE — which may have cosmological significance.

Third, in the classical limit, the entropic field amplitude is *frozen*: the field $S(r)$ is a classical background configuration, not a quantum operator subject to fluctuations. The path integral over entropic field configurations is dominated by a single saddle point — the classical solution — and quantum corrections (loop diagrams in the entropic field theory) are suppressed by powers of $\hbar$. Under these three conditions — negligible SOA corrections, vanishing OCI, and frozen amplitude — the full GEFE reduce exactly to the vacuum Einstein equations, $G_{\mu\nu} = 0$, or equivalently $R_{\mu\nu} = 0$. It is in this limit that the Schwarzschild solution emerges.

We emphasize that the classical limit is the regime in which the Theory of Entropicity makes the *same* predictions as general relativity. Beyond this limit — at short distances, high curvatures, or quantum scales — the ToE predicts corrections that could be observationally tested. The Schwarzschild derivation presented here is a *consistency check*, not the frontier of the theory's predictive power. The frontier lies in the corrections: the entropic hair, the dark curvature, the quantum modifications to the horizon structure, and the resolution of the classical singularity. These are addressed in subsequent appendices and Letters.

3. Step 1: Radial Entropic Field Equation (MEE)

3.1. The Minimal Entropic Equation in the Vacuum, Minimal-Coupling Limit

The derivation of the Schwarzschild solution from the Theory of Entropicity begins with the fundamental field equation governing the entropic field $S(x)$. This is the *Minimal Entropic Equation* (MEE), the Euler-Lagrange equation obtained by varying the Obidi Action with respect to the entropic field. The MEE is to the Theory of Entropicity what the Klein-Gordon equation is to scalar field theory, or what the Maxwell equations are to electrodynamics: it is the dynamical equation of motion for the fundamental field of the theory.

The full MEE, as derived in Letter IE, contains the covariant d'Alembertian of the entropic field, the derivative of the entropic potential, a non-minimal coupling term involving the Ricci scalar and the entropic field, and a source term encoding external matter contributions. In the most general form, the MEE reads $\alpha \square S + V'(S) - \beta R'_{ent}(S) = J(x)$, where α and β are coupling constants, $R'_{ent}(S)$ is the entropic curvature contribution from non-minimal coupling, and $J(x)$ is the external source. Under the three assumptions stated in Section 2 — static spherical symmetry, minimal coupling ($\xi \rightarrow 0$), and the classical vacuum regime ($J(x) = 0$) — the MEE simplifies dramatically.

In the minimal coupling limit and in vacuum (no external matter), the entropic field S satisfies the Minimal Entropic Equation (MEE) of the form

(IE-S.100)

$$\square S - V(S) = 0$$

where $\square$ is the d'Alembertian with respect to the emergent spacetime metric, and $V(S)$ is the entropic potential.

Equation (IE-S.100) is the cornerstone of the present derivation. Let us unpack its content carefully. The d'Alembertian operator $\square$ is the covariant generalization of the wave operator (the Laplacian in the case of a static field) and is defined in terms of the spacetime metric $g_{\mu\nu}$ by

$$\square S = (1/\sqrt{-g}) \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu S)$$

where $g = \det(g_{\mu\nu})$ is the determinant of the metric tensor. For a static field $S = S(r)$ in a static, spherically symmetric background, the time derivatives vanish, and the d'Alembertian reduces to the radial part of the covariant Laplacian. The structure of this operator depends on the background metric, but in the weak-field regime — where the background is approximately flat — the d'Alembertian reduces to the flat-space radial Laplacian.

The entropic potential $V(S)$ encodes the self-interaction of the entropic field. It plays a role analogous to the Higgs potential in the Standard Model of particle physics or the inflaton potential in inflationary cosmology: it

determines the vacuum structure of the theory (the values of S at which $V'(S) = 0$, which correspond to equilibrium configurations of the entropic field) and the mass spectrum of small fluctuations around those vacua. Near the entropic vacuum $S = 0$ (or, more precisely, near the asymptotic background value S_0), the potential can be Taylor-expanded as

$$V(S) = V_0 + \frac{1}{2} m^2 S^2 + (\text{higher-order terms})$$

where V_0 is a cosmological constant contribution (irrelevant for the local Schwarzschild problem), m is the *entropic mass scale* (the mass of the entropic field quantum), and the higher-order terms encode nonlinear self-interactions. The entropic mass m sets the fundamental length scale $1/m$ (in natural units) beyond which the entropic field is exponentially suppressed — the *Compton wavelength* of the entropic quantum. The derivative of the potential, which appears in (IE-S.100), is therefore $V'(S) = m^2 S + O(S^2)$ near the vacuum.

3.2. Reduction to the Radial Equation

The reduction of the MEE (IE-S.100) to a radial ordinary differential equation proceeds through the assumption of static spherical symmetry (Assumption 1) and the weak-field approximation (Assumption 3). For a static field $S = S(r)$ in an approximately flat background, the d'Alembertian reduces to the flat-space Laplacian in spherical coordinates:

$$\square S \rightarrow \nabla^2 S = (1/r^2) d/dr(r^2 dS/dr)$$

This is the standard radial Laplacian in three spatial dimensions. The factor r^2 in the derivative reflects the geometry of three-dimensional Euclidean space: the surface area of a sphere of radius r is $4\pi r^2$, and the r^2 factor ensures that the Laplacian accounts for the divergence of the radial flux

across spherical surfaces. Substituting this form of the Laplacian and the linearized potential $V'(S) = m^2 S$ into (IE-S.100), we obtain the radial entropic field equation.

In the weak-field, static, spherically symmetric regime, we approximate the background as effectively flat and write the radial equation as

(IE-S.101)

(1/

r

2

)

d

/

dr

(

r

2

dS

/

dr

)

–

m

2

S

= 0

where m is an effective entropic mass scale (coming from $V(S) \approx \frac{1}{2} m^2 S^2$ near $S = 0$).

Equation (IE-S.101) is a well-known equation in mathematical physics, appearing in diverse contexts under different names. In the mathematical literature, it is the *modified spherical Bessel equation* (or the radial part of the *modified Helmholtz equation* $(\nabla^2 - m^2)S = 0$ in spherical coordinates for the $l = 0$ angular momentum sector). In nuclear physics, it is the equation whose solutions yield the Yukawa potential — the potential mediating the strong nuclear force between nucleons via meson exchange, as proposed by Hideki Yukawa in 1935. In condensed matter physics, it describes the screened Coulomb potential (the Debye-Hückel potential) in an electrolyte, where the screening length $1/m$ is the Debye length. In all these contexts, the physical content is the same: the mass parameter m introduces a length scale $1/m$ beyond which the field is exponentially suppressed.

To solve (IE-S.101), we make the standard substitution $S(r) = u(r)/r$, which transforms the equation into $u'' - m^2 u = 0$ (where primes denote derivatives with respect to r). This is a linear, constant-coefficient ODE whose general solution is $u(r) = Ae^{-mr} + Be^{+mr}$. The regularity condition at $r \rightarrow \infty$ (the requirement that the field not grow exponentially at large distances, which would represent an unphysical source at infinity) forces $B = 0$. Restoring the factor of $1/r$, we obtain the Yukawa solution.

3.3. The Yukawa Solution and Its Physical Interpretation

The general static, spherically symmetric solution of (IE-S.101) is the Yukawa form

(IE-S.102)

$$S_0 + \frac{S_1}{r} \exp(-mr)$$

where S_0 and S_1 are integration constants.

Let us discuss the physical interpretation of each element of this solution. The constant S_0 is the *asymptotic entropic background* — the value of the entropic field at spatial infinity, $S(r \rightarrow \infty) = S_0$. In the ToE framework, S_0 represents the ambient entropic density of the universe far from any localized gravitating body. It is the entropic analogue of the cosmological background: just as the cosmic microwave background provides a thermal background temperature $T_{CMB} \approx 2.725$ K pervading all of space, the entropic background S_0 provides a baseline information density from which local concentrations (gravitating bodies) emerge as perturbations.

The constant S_I is the *entropic charge* — the "strength" of the entropic source. It is determined by the total entropic content of the central body and plays a role in the ToE framework analogous to the role of the mass M in Newtonian gravity or the electric charge Q in electrostatics. As we shall show in Step 4, the entropic charge S_I is directly proportional to the Schwarzschild mass parameter M , establishing the identification of mass as an emergent entropic quantity.

The exponential factor $\exp(-mr)$ produces *Yukawa screening*: at distances $r \gg 1/m$, the entropic field decays exponentially faster than the Coulomb-like $1/r$ falloff, effectively confining the entropic influence to a finite range. The screening length $1/m$ is the *Compton wavelength* of the entropic quantum (in natural units). If the entropic mass m is non-zero, the gravitational interaction mediated by the entropic field has a finite range — a Yukawa-type modification of gravity. Such modifications are tightly constrained by solar-system tests and laboratory experiments (Eöt-Wash torsion balance experiments, for example, constrain Yukawa deviations from Newtonian gravity at sub-millimeter scales), but they remain a live possibility at galactic and cosmological scales, where they could contribute to modified gravity phenomenology.

We can verify by direct substitution that (IE-S.102) satisfies (IE-S.101). Setting $f(r) = (S_1/r)e^{-mr}$, we compute $f'(r) = S_1e^{-mr}(-1/r^2 - m/r)$ and $r^2f'(r) = S_1e^{-mr}(-1 - mr)$. Taking the derivative: $d/dr[r^2f'] = S_1e^{-mr}(m + m^2r - m) = m^2S_1re^{-mr}$. Dividing by r^2 gives $(1/r^2)d/dr[r^2f'] = m^2f(r)$, confirming that (IE-S.101) is satisfied. The constant S_0 satisfies the equation trivially since its derivatives vanish (the mass term m^2S_0 is absorbed into a redefinition of the potential or, equivalently, into the cosmological constant sector).

3.4. The Massless Limit: Coulomb-Like Entropic Profile

For the gravitational sector in the long-range, classical limit, we take the massless limit $m \rightarrow 0$, which yields the Coulomb-like entropic profile

(IE-S.102.1)

S

(

r

) =

S

0

+

S

1

/

r

This $S(r)$ will serve as the prototype entropic configuration associated with a spherically symmetric mass distribution.

The massless limit $m \rightarrow 0$ is the regime in which the entropic interaction becomes *long-range*, precisely like the gravitational and electromagnetic interactions. This is physically natural: gravity is a long-range force (there is no gravitational screening, no gravitational analogue of the Debye-Hückel effect at macroscopic scales), and this long-range character is reflected in the $1/r$ falloff of the Coulomb-like profile (IE-S.102.1). The massless limit is therefore the gravitational limit of the entropic field theory.

The Coulomb-like profile $S(r) = S_0 + S_I/r$ is the entropic analogue of the Newtonian gravitational potential $\Phi(r) = -GM/r$. The $1/r$ falloff is not imposed by hand but is *dictated by three-dimensional geometry*: it is the unique solution of Laplace's equation $\nabla^2 S = 0$ that is spherically symmetric, regular at infinity, and singular at the origin (corresponding to a point source). This is the content of Gauss's law applied to the entropic flux: the total entropic flux through any sphere enclosing the source is proportional to the enclosed entropic charge S_I , and the flux per unit area scales as $1/r^2$, which requires the field to scale as $1/r$.

The identification $S_I \sim M$ (to be made precise in Step 4) reveals a deep structural parallel. Just as the electrostatic potential $\Phi_E(r) = Q/(4\pi\epsilon_0 r)$ is determined by the electric charge Q , and just as the Newtonian gravitational potential $\Phi(r) = -GM/r$ is determined by the mass M , the entropic profile S_I/r is determined by the entropic charge S_I . In all three cases, the $1/r$ spatial dependence is universal (it follows from the three-dimensional Gauss's law),

and the physical content resides entirely in the coefficient — the charge, the mass, or the entropic charge. The Theory of Entropicity unifies these by identifying mass as entropic charge: the mass of a gravitating body *is* the strength of its entropic field.

It is worth pausing to appreciate the logical structure. We began with the Minimal Entropic Equation — the field equation of the entropic field, derived from the Obidi Action — and by imposing three physically transparent assumptions (static spherical symmetry, minimal coupling, classical limit), we obtained the Coulomb-like profile (IE-S.102.1). This profile is not assumed; it is *derived*. The entropic field around a static, spherically symmetric source is necessarily of the form $S_0 + S_1/r$. This is the starting point for the construction of the emergent spacetime geometry.

4. Step 2: Fisher-Rao Information Metric for $S(r)$

4.1. From the Entropic Field to Information Geometry: The Conceptual Bridge

The transition from the entropic field (a scalar function defined on the spatial manifold) to a metric tensor (a geometric structure defining distances, angles, and curvature) is the conceptual heart of the Theory of Entropicity. This transition is mediated by the *Fisher-Rao information metric* — the unique, natural, diffeomorphism-invariant metric on the space of probability distributions. The Fisher-Rao construction is the bridge between the entropic substrate (the information manifold M_I) and the emergent spacetime (the

physical manifold M_S), and it is through this bridge that the entropic field $S(r)$ acquires geometric significance.

The key idea, developed at length in Letter IE, is as follows. The entropic field $S(x)$ at each point x of the spatial manifold induces a probability distribution on the entropic substrate — a statistical description of the microscopic degrees of freedom at that point. As one moves from point to point, the entropic field changes, and the induced probability distribution changes accordingly. The space of all these distributions, parameterized by the spatial coordinates, forms a *statistical manifold*, and the natural metric on this manifold — the Fisher-Rao metric — endows it with a Riemannian geometry. This geometry, the *information geometry* of the entropic substrate, is what the emergence map Φ converts into physical spacetime geometry.

The theoretical foundations of this construction rest on three pillars. First, the work of Ronald Aylmer Fisher (1925), who introduced the concept of *Fisher information* as a measure of the information content of a statistical sample. Second, the work of Calyampudi Radhakrishna Rao (1945), who recognized that the Fisher information matrix defines a Riemannian metric on the space of parameterized probability distributions. Third, the theorem of Nikolai Nikolaevich Čencov (1972, published in full in 1982), which establishes that the Fisher-Rao metric is the *unique* Riemannian metric on a statistical manifold that is invariant under all sufficient statistics — i.e., under all transformations of the data that preserve the statistical information. The Čencov uniqueness theorem is of profound significance for the Theory of Entropicity: it means that the information metric is not a choice but a *necessity*. Any diffeomorphism-invariant geometric structure constructed from a family of probability distributions must be proportional to the Fisher-Rao metric. The emergent spacetime geometry is therefore uniquely determined (up to a constant) by the entropic field.

4.2. The Gaussian Statistical Model

To connect the entropic field to geometry, we consider a one-parameter family of probability distributions $p(x / r)$ whose macroscopic parameter is controlled by $S(r)$. For concreteness, take a simple Gaussian family where the variance $\sigma^2(r)$ is proportional to $S(r)$:

(IE-S.102.2)

σ

2

(

r

)

$\propto$

S

(

r

)

The choice of a Gaussian statistical model is made for concreteness and computational transparency, but it is important to emphasize that the qualitative result — a Fisher-Rao metric proportional to $[S'/S]^2$ — is *not* an artifact of the Gaussian assumption. For any member of the exponential family (Gaussian, Poisson, exponential, gamma, etc.), the Fisher-Rao metric depends on the parameter of interest through the logarithmic gradient of the

parameter, and the resulting information metric has the same structural form. The Čencov uniqueness theorem guarantees this universality.

The proportionality $\sigma^2(r) \propto S(r)$ has a natural physical interpretation in the ToE framework. The variance σ^2 measures the spread of the microscopic degrees of freedom around their mean — it quantifies the "width" of the statistical distribution of the entropic substrate at each spatial point. The proportionality to $S(r)$ means that regions of higher entropic density have greater statistical spread: more entropy corresponds to more microscopic configurations, more fluctuation, more uncertainty. This is physically natural and consistent with the thermodynamic intuition that entropy measures the logarithm of the number of accessible microstates. Near the central gravitating body, where $S(r)$ is large, the microscopic degrees of freedom are highly spread out (large variance); far from the body, where $S(r) \approx S_0$, the distribution is narrow (small variance). The Fisher-Rao metric, which measures the "statistical distinguishability" of nearby distributions, will therefore be large where the entropic field changes rapidly (near the source) and small where it is nearly constant (at large distances).

4.3. Definition of the Fisher-Rao Metric

For a one-parameter family $p(x / \theta)$, the Fisher-Rao metric is defined by

(IE-S.102.3)

g

|

(

$$\theta$$

$$)=$$

$$E$$

$$[($$

$$\partial$$

$$\ln$$

$$p$$

$$/$$

$$\partial$$

$$\theta$$

$$)$$

2

$$]$$

and in the present case the relevant parameter is $\theta = r$.

The Fisher-Rao metric (IE-S.102.3) has a beautiful information-theoretic interpretation. The quantity $\partial \ln p / \partial \theta$ is the *score function* — the logarithmic sensitivity of the probability distribution to changes in the parameter θ . Its expectation value vanishes ($E[\partial \ln p / \partial \theta] = 0$ for any regular statistical model), so the Fisher-Rao metric is the variance of the score function. It measures how "statistically distinguishable" two nearby distributions $p(x / \theta)$ and $p(x / \theta + d\theta)$ are: a large Fisher information means that a small change in θ produces a large change in the statistical model, making the two distributions easy to

distinguish. The Fisher-Rao metric thereby converts statistical distinguishability into geometric distance.

The connection to the Kullback-Leibler divergence makes this precise: the Fisher-Rao metric is the Hessian of the KL-divergence, $g_I(\theta) = \partial^2 D_{KL}(p(\cdot/\theta) // p(\cdot/\theta'))/\partial\theta^2$ evaluated at $\theta' = \theta$. The KL-divergence measures the information lost when one approximates the true distribution $p(\cdot/\theta)$ with a nearby distribution $p(\cdot/\theta')$, and the Fisher-Rao metric is the leading-order term in this loss. The information geometry of the entropic substrate is therefore a geometry of *information loss*: the distance between two points on the information manifold measures how much information is lost in confusing the corresponding distributions.

4.4. Explicit Computation for the Gaussian Model

A straightforward computation for a Gaussian with variance $\sigma^2(r)$ gives a Fisher-Rao metric component of the form

(IE-S.102.4)

g

l,rr

(

r

) =

C

.

[
S
'(
r
)]
2
/[
S
(
r
)]
2

where C is a positive constant depending on the normalization of the statistical model, and $S'(r) = dS/dr$.

Let us present the full computation explicitly. Consider the Gaussian probability density

$$p(x | r) = (1/\sqrt{2\pi\sigma^2(r)}) \exp(-x^2/(2\sigma^2(r)))$$

with $\sigma^2(r) = \kappa S(r)$ for some positive proportionality constant κ . The log-likelihood is

$$\ln p = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - x^2/(2\sigma^2)$$

Differentiating with respect to r (using $d\sigma^2/dr = \kappa S'(r)$ and $d \ln \sigma^2/dr = S'(r)/S(r)$):

$$\partial \ln p/\partial r = -\frac{1}{2}(S'/S) + (x^2/(2\sigma^4))(\kappa S) = (S'/S)[-1/2 + x^2/(2\sigma^2)]$$

Squaring and taking the expectation (using the Gaussian moments $E[x^2/\sigma^2] = 1$ and $E[x^4/\sigma^4] = 3$):

$$E[(\partial \ln p/\partial r)^2] = (S'/S)^2 \cdot E[(-1/2 + x^2/(2\sigma^2))^2] = (S'/S)^2 \cdot (1/4 - 1/2 + 3/4) = \frac{1}{2}(S'/S)^2$$

This confirms (IE-S.102.4) with $C = 1/2$ for the Gaussian model. The result has a remarkable structural property: the Fisher-Rao metric depends on the *logarithmic gradient* $S'/S = d \ln S/dr$ of the entropic field, not on the field itself. This is deeply connected to the information-theoretic principle that only *relative* changes in entropy carry geometric meaning. An additive shift in $S(r)$ does not change the logarithmic gradient (provided $S(r) > 0$), and thus does not change the Fisher-Rao metric. The information geometry is insensitive to the overall "level" of the entropy — only the *spatial variation* of the entropy, measured in relative terms, is geometrically relevant.

For alternative statistical models, the constant C changes but the structural form $[S'/S]^2$ is preserved. For instance, for a Poisson model with mean $\lambda(r) \propto S(r)$, the Fisher-Rao metric is $g_{l,r} = [S'/S]^2/S$, which for large S scales as $[S'/S]^2$. For a general exponential family model with natural parameter $\eta(r)$ related to $S(r)$, the Fisher-Rao metric involves the second derivative of the log-partition function, but in the regime of interest the dependence on S'/S is universal. This universality, guaranteed by the Čencov theorem, ensures that the qualitative features of the emergent geometry do not depend on the specific statistical model used to define the information metric.

4.5. Evaluation for the Coulomb-Like Entropic Profile

For the Coulomb profile (IE-S.102.1),

(IE-S.102.5)

S

(

r

) =

S

0

+

S

1

/

r

we have

(IE-S.102.6)

S

'(

$$\frac{r_1 - r_2}{r_1} =$$

and therefore

(IE-S.102.7)

$$g(l, r) = \frac{r_1 - r_2}{r_1} \propto \frac{S_1 - S_2}{S_1}$$

/
 r
 4
)/[
 S
 0
 +
 S
 1
 /
 r
]
 2

The behavior of $g_{l,rr}(r)$ at different radial scales is highly informative and reveals the geometric structure of the information manifold.

At large distances ($r \gg S_l/S_0$): The denominator is dominated by S_0^2 , and the Fisher-Rao metric becomes $g_{l,rr} \propto S_l^2/(S_0^2 r^4)$, which falls off rapidly with distance. The information metric becomes flat at large distances — the information manifold is asymptotically Euclidean, reflecting the fact that far from the gravitating body, the entropic field is nearly constant and the statistical manifold is nearly uniform. This corresponds to the asymptotic flatness of the Schwarzschild spacetime.

At small distances ($r \ll S_I/S_0$): The denominator is dominated by $(S_I/r)^2 = S_I^2/r^2$, and the Fisher-Rao metric becomes $g_{I,rr} \propto 1/r^2$, which diverges as $r \rightarrow 0$. The information manifold has a curvature singularity at the origin, corresponding to the central mass. This singularity of the Fisher-Rao metric is the information-geometric precursor of the curvature singularity of the Schwarzschild spacetime at $r = 0$.

At the transition scale $r \sim r_* = S_I/S_0$: The two terms in the denominator are comparable, and the Fisher-Rao metric transitions between the two asymptotic regimes. This transition scale is the entropic analogue of the Schwarzschild radius — the characteristic scale at which the entropic field perturbation S_I/r becomes comparable to the background S_0 . We shall see in Step 4 that $r_* = S_I/S_0$ is related to the Schwarzschild radius $r_s = 2GM/c^2$ through the mass-entropy identification $M = \alpha S_I$.

One can compute the Ricci scalar of the one-dimensional information metric $g_{I,rr}(r) dr^2$ in the radial sector to verify that the information manifold is indeed curved. In one dimension, the Ricci scalar of a metric $g(r) dr^2$ is related to $g(r)$ by a formula involving its second logarithmic derivative. The non-vanishing of this curvature confirms that the information manifold is not flat — it has a genuine geometric structure that, via the emergence map, becomes the spacetime curvature of the Schwarzschild geometry.

4.6. The Full Information Manifold Metric

The angular components of the information metric are taken to respect spherical symmetry, so that the information manifold metric in the radial-angular sector can be written as

(IE-S.102.8)

ds

l

2

$=$

g

l, rr

$($

r

$)$

dr

2

$+$

r

2

$($

$d\theta$

2

$+ \sin$

2

θ

$d\phi$

2

)

The exact overall normalization of $g_{l,rr}$ is not essential here; what matters is that the information manifold is spherically symmetric and that its radial component is determined by the entropic profile $S(r)$.

The metric (IE-S.102.8) defines a three-dimensional Riemannian manifold with the topology of $\mathbb{R}^3 \setminus \{0\}$ (three-dimensional Euclidean space with the origin removed, reflecting the point-source singularity at $r = 0$). The angular part $r^2(d\theta^2 + \sin^2\theta d\varphi^2)$ is the standard metric on the two-sphere of radius r , ensuring that the area of the sphere at coordinate radius r is $4\pi r^2$. This is the *areal radius* gauge — a coordinate choice that is both natural and computationally convenient.

It should be emphasized that (IE-S.102.8) is the *spatial section* of the information manifold at a fixed time. The full four-dimensional information metric includes a temporal component that encodes the causal structure of the information manifold. However, in the static case considered here, the temporal and spatial sectors decouple, and the spatial metric (IE-S.102.8) contains all the information needed for the spatial part of the emergence map. The temporal component of the physical spacetime metric will be determined separately, in Step 3, by the dynamical equations (the vacuum Einstein equations in the classical limit).

The information manifold (M_l, g_l) defined by (IE-S.102.8) is the "statistical shadow" of the entropic field — the geometric structure that encodes how the entropic probability distributions vary from point to point in space. Near the central body, where the entropic field changes rapidly, the Fisher-Rao

distances are large — the distributions at nearby radii are highly distinguishable. Far from the body, where the field is nearly constant, the Fisher-Rao distances are small — the distributions are nearly indistinguishable. The information manifold is therefore "highly curved" near the source and "nearly flat" at large distances, mirroring the behavior of the Schwarzschild spacetime.

5. Step 3: Emergence Map and Physical Spacetime Metric

5.1. The Emergence Map: From Information Geometry to Physical Spacetime

With the information manifold metric (IE-S.102.8) in hand, we now invoke the central mechanism of the Theory of Entropicity: the *emergence map* $\Phi: M_I \rightarrow M_S$, which identifies the physical spacetime metric with the Fisher-Rao information metric. The emergence map was established in Letter IE as the fundamental link between the information manifold (M_I, g_I) and the physical spacetime (M_S, g_S) . It is not a coordinate transformation or a gauge choice; it is a *constitutive relation* — a statement about the ontological nature of spacetime geometry within the ToE framework. The emergence map asserts that spacetime geometry is not fundamental but *emergent*: it is generated by the entropic field through the information metric.

The emergence map identifies the physical spacetime metric g_S with the Fisher-Rao information metric g_I up to a constant factor λ :

(IE-S.102.9)

g

$S_{,\mu\nu}$

$=$

λ

g

$I_{,\mu\nu}$

with λ a constant in the classical regime.

The conformal relation (IE-S.102.9) is the simplest form of the emergence map, valid in the classical regime where quantum corrections to the emergence constant are negligible. As derived in Letter IE, the emergence constant is

$$\lambda = I_P^2/(4k_B) = \hbar G/(4k_B c^3)$$

where $I_P = (\hbar G/c^3)^{1/2}$ is the Planck length, k_B is Boltzmann's constant, $\hbar$ is the reduced Planck constant, G is Newton's gravitational constant, and c is the speed of light. The emergence constant λ has dimensions of (length)²/(information), which is necessary for dimensional consistency: the information metric g_I has dimensions of (information)² (since it is defined in terms of log-likelihoods, which are dimensionless in natural units or have dimensions of information in information-theoretic units), and the physical metric g_S has dimensions of (length)². The constant λ bridges these dimensions.

The physical significance of the emergence map cannot be overstated. It is the mathematical realization of the central thesis of the Theory of Entropicity: *spacetime geometry is generated by the entropic field through the information metric*. The information manifold is not a mathematical convenience or an analogy; it is the *substrate* from which physical spacetime emerges. The emergence map Φ is the mechanism of this emergence, and the constant λ sets the scale at which informational structure becomes geometric structure.

The Curvature Transfer Theorem (Theorem 3.1 of Letter IE) guarantees that the Riemann curvature tensor of the physical spacetime (M_S, g_S) is determined by the pushforward of the Riemann curvature tensor of the information manifold (M_I, g_I) . Specifically, $R^S_{\mu\nu\rho\sigma} = \lambda \Phi_*(R^I_{\mu\nu\rho\sigma}) + O(1/N)$, where N is the number of entropic degrees of freedom. In the classical limit $N \rightarrow \infty$, the correction term vanishes, and the curvature transfer is exact. The Schwarzschild curvature is therefore *information curvature*, rescaled by λ .

5.2. The Physical Spacetime Metric Ansatz

For a static, spherically symmetric spacetime, we write the physical metric in standard Schwarzschild-like coordinates as

(IE-S.102.10)

ds

S

2

=

-

A

(

r

)

c

2

dt

2

+

B

(

r

)

dr

2

+

r

2

(

dθ

2

+ sin

2

θ

$d\varphi$

2

)

This is the most general static, spherically symmetric line element compatible with the symmetry assumptions. The metric is characterized by two unknown functions of the radial coordinate: $A(r)$, the *temporal metric function* (also called the "lapse function" in the ADM formalism or the "redshift function" in the astrophysical literature), and $B(r)$, the *radial metric function* (related to the proper radial distance). The angular part $r^2(d\theta^2 + \sin^2\theta d\varphi^2) = r^2 d\Omega^2$ is already in the standard form on the two-sphere, corresponding to the choice of *areal radius* r (the coordinate is defined so that the area of the sphere at coordinate radius r is $4\pi r^2$). This gauge choice eliminates one functional degree of freedom and simplifies the Einstein equations considerably.

The function $A(r)$ determines the gravitational redshift and time dilation experienced by static observers at radius r . A photon emitted at radius r and received at infinity is redshifted by the factor $\sqrt{A(r)}$, and a clock at radius r runs slow relative to a clock at infinity by the same factor. The function $B(r)$ determines the proper radial distance: the proper distance between two radial shells at r_1 and r_2 is $\int_{r_1}^{r_2} \sqrt{B(r)} dr$, which differs from the coordinate distance $r_2 - r_1$ whenever $B(r) \neq 1$.

Asymptotic flatness — the requirement that spacetime become Minkowskian at large distances from the source — imposes the boundary conditions $A(r) \rightarrow 1$ and $B(r) \rightarrow 1$ as $r \rightarrow \infty$. These conditions ensure that the metric reduces to the flat Minkowski metric $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$ far from the gravitating body, as required by the physical situation (isolated body in otherwise empty space).

5.3. Constraints from the Emergence Map

The emergence map (IE-S.102.9) implies that, in the radial-angular sector,

(IE-S.102.11)

B

(

r

)

$\propto$

g

l,rr

(

r

)

and the angular part is already in the standard form $r^2 d\Omega^2$. The time component $A(r)$ is determined dynamically by the GEFE/OFE in the vacuum.

The proportionality (IE-S.102.11) is the key constraint imposed by the emergence map on the spatial part of the physical metric. The radial metric function $B(r)$ of the physical spacetime inherits its radial dependence directly from the Fisher-Rao metric $g_{lr}(r)$ of the information manifold. This is a concrete realization of the ToE thesis: the spatial geometry of the physical spacetime is *determined by* the information geometry of the entropic substrate.

The temporal metric function $A(r)$, however, is *not* directly determined by the information metric. This is because the Fisher-Rao metric is a *Riemannian* (positive-definite) metric on the statistical manifold, while the physical spacetime metric is *Lorentzian* (indefinite, with signature $(-, +, +, +)$). The Lorentzian signature introduces causal structure — the distinction between timelike, spacelike, and null directions — which has no direct counterpart in the Riemannian information metric. The temporal component of the physical metric, which determines the causal structure, must therefore be fixed by additional dynamical information: specifically, by the field equations of the theory (the GEFE in the full framework, or the vacuum Einstein equations in the classical limit).

This asymmetry between the spatial and temporal components is physically natural. The spatial geometry reflects the *static* configuration of the entropic field — it is a snapshot of the information content at a given time. The temporal geometry reflects the *dynamics* — the way the information content evolves in time (or, in the static case, the way the causal structure is constrained by the field equations). The emergence map provides the spatial geometry directly; the dynamics provides the temporal geometry.

5.4. The Classical Limit: Vacuum Einstein Equations

In the classical, large-scale limit, the GEFE reduce to the vacuum Einstein equations

(IE-S.102.12)

R

$\mu\nu$

$-$

$($

$\frac{1}{2}$

$)$

R

g

$\mu\nu$

$= 0$

The reduction of the full GEFE to (IE-S.102.12) proceeds through the hierarchy of limits described in Assumption 3. Let us recapitulate and elaborate. The full GEFE, as derived in Letter IE, have the schematic form

$$G_{\mu\nu} = (8\pi G/c^4)[T_{\mu\nu}^{\text{LOA}} + T_{\mu\nu}^{\text{SOA}} + T_{\mu\nu}^{\text{OCI}}]$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor, $T_{\mu\nu}^{\text{LOA}}$ is the Leading-Order Approximation stress-energy (containing the kinetic energy $\partial_\mu S \partial_\nu S$, the potential $V(S)$, and the non-minimal coupling ξRS^2), $T_{\mu\nu}^{\text{SOA}}$ contains the

Second-Order Approximation corrections (higher-curvature terms from the Seeley-DeWitt expansion, suppressed by $(l_P/L)^2$), and $T_{\mu\nu}^{OCI}$ is the Obidi Curvature Invariant tensor (encoding the curvature mismatch between M_I and M_S).

In the classical limit:

5. **SOA corrections are negligible:** The higher-curvature corrections are suppressed by $(l_P/L)^2 \sim 10^{-70}$ for astrophysical scales, rendering them utterly undetectable.
6. **OCI vanishes:** When the emergence map is a conformal isometry with constant λ , the curvature mismatch between M_I and M_S vanishes identically, and $T_{\mu\nu}^{OCI} = 0$.
7. **LOA vanishes in vacuum with minimal coupling:** With $\xi = 0$, the LOA stress-energy involves only $\partial_\mu S \partial_\nu S$ and $V(S)$. In the vacuum regime, the entropic field is a classical background that has already been absorbed into the emergent metric via the Fisher-Rao construction. The effective stress-energy of the entropic field, after the emergence map has been applied, is already encoded in the geometry of g_S and does not appear as an independent source term.

What remains is the vacuum Einstein equation $G_{\mu\nu} = 0$, which is precisely (IE-S.102.12). Taking the trace of (IE-S.102.12) with the metric $g^{\mu\nu}$ gives $-R = 0$ (in four dimensions, $g^{\mu\nu}g_{\mu\nu} = 4$, so the trace is $R - 2R = -R$), confirming that the Ricci scalar vanishes: $R = 0$. The vacuum Einstein equations therefore simplify to the Ricci-flat condition:

$$R_{\mu\nu} = 0$$

This is the definitive form of the vacuum field equations, and it is this condition — Ricci-flatness — that uniquely determines the Schwarzschild geometry, by Birkhoff's theorem.

5.5. Solution: The Schwarzschild Geometry

For the static, spherically symmetric ansatz (IE-S.102.10), these equations yield the standard system of ordinary differential equations for $A(r)$ and $B(r)$. Solving (IE-S.102.12) under the requirement of asymptotic flatness ($A(r) \rightarrow 1$, $B(r) \rightarrow 1$ as $r \rightarrow \infty$) gives the unique Schwarzschild solution

(IE-S.102.13)

A

(

r

) = 1

–

2

GM

/(

rc

2

)

(IE-S.102.14)

B

(

r

) = [1

-

2

GM

/(

rc

2

)]

-1

The derivation of (IE-S.102.13)–(IE-S.102.14) from the Ricci-flat condition $R_{\mu\nu} = 0$ is a standard exercise in general relativity, but its logical structure merits a brief review. For the metric ansatz (IE-S.102.10), the non-vanishing Christoffel symbols involve A , A' , B , B' (primes denoting d/dr), and the Ricci tensor components yield three independent equations:

The (t,t) component: $R_{tt} = 0$ gives an equation involving A'' , A' , B' , and A , B .

The (r,r) component: $R_{rr} = 0$ gives a second equation involving the same quantities.

The (θ, θ) component: $R_{\theta\theta} = 0$ gives a third equation that serves as a consistency condition.

A crucial intermediate result emerges from the combination of the (t, t) and (r, r) equations: $A(r)B(r) = \text{const.}$ The asymptotic flatness condition $A \rightarrow 1, B \rightarrow 1$ as $r \rightarrow \infty$ fixes this constant to unity:

$$A(r) B(r) = 1$$

This is the fundamental constraint relating the temporal and radial metric functions of the Schwarzschild geometry: they are reciprocals of each other. Substituting $B = 1/A$ into the (θ, θ) equation yields a first-order ODE for $A(r)$:

$$rA' + A = 1 \quad \Rightarrow \quad d(rA)/dr = 1 \quad \Rightarrow \quad A(r) = 1 - C/r$$

where C is an integration constant. Matching to the Newtonian limit (where $A(r) \approx 1 + 2\Phi/c^2 = 1 - 2GM/(rc^2)$) identifies $C = 2GM/c^2 = r_s$, the Schwarzschild radius. This completes the derivation of (IE-S.102.13)–(IE-S.102.14).

The Schwarzschild radius $r_s = 2GM/c^2$ marks the event horizon — the boundary of the black hole region from which no signal can escape to infinity. For a solar-mass object, $r_s \approx 3$ km; for the Earth, $r_s \approx 9$ mm. At $r = r_s$, the coordinate singularity in $B(r)$ is an artifact of the Schwarzschild coordinates and can be removed by choosing different coordinates (Eddington-Finkelstein, Kruskal-Szekeres, Painlevé-Gullstrand). The true curvature singularity at $r = 0$, where the Kretschner scalar $K = 48G^2M^2/(c^4r^6)$ diverges, is coordinate-independent and signals a genuine breakdown of the classical theory.

5.6. Summary of Step 3

Thus, in the classical limit where the GEFE reproduce the vacuum Einstein equations, the emergent spacetime metric generated by a static, spherically symmetric entropic configuration is necessarily the Schwarzschild metric.

This result is the central achievement of this appendix. The Schwarzschild metric is not *assumed* in the Theory of Entropicity — it *emerges* from the entropic field configuration through a precisely defined chain of constructions: the entropic field $S(r)$ induces a statistical manifold; the Fisher-Rao metric on this manifold defines the information geometry; the emergence map converts information geometry into physical spacetime geometry; and the vacuum Einstein equations (the classical limit of the GEFE) constrain the temporal component, completing the Schwarzschild metric. Each link in this chain is mathematically rigorous and physically motivated.

The result validates the GEFE/OFE as a viable theory of gravity: in the regime where they must agree with GR (the classical, weak-field, large-scale limit), they do agree — exactly. The Schwarzschild solution is not lost or modified in the entropic reformulation; it is recovered intact. This is a non-trivial consistency check, because the GEFE have a very different structure from the Einstein equations (they include SOA corrections, the OCI tensor, and the non-minimal coupling), and it is not *a priori* obvious that they reduce to the correct classical limit. The fact that they do is a powerful validation of the theoretical framework.

6. Step 4: Identification of the Mass Parameter in Terms of Entropic Data

6.1. The Significance of the Mass Identification

The remaining task is to relate the Schwarzschild mass parameter M to the entropic field parameters S_0 and S_I .

This identification is arguably the most profound result of this appendix, for it addresses a question that general relativity leaves entirely unanswered: *What is mass?* In Einstein's theory, the mass M appearing in the Schwarzschild metric is a *free parameter* — it is set by initial conditions and has no deeper explanation within the theory. The Einstein equations tell us *how* mass curves spacetime, but they do not tell us *what* mass is or *why* it curves spacetime. Mass is an input, not an output, of the theory.

In the Theory of Entropicity, this situation changes fundamentally. The mass M is not a free parameter but a *derived quantity* — it is determined by the entropic field configuration, specifically by the entropic charge S_I . The mass of a gravitating body is not a primitive datum but an emergent consequence of the entropic field that the body generates. This represents a genuine explanatory advance: the ToE explains *what mass is* (an entropic charge — a measure of the total information content concentrated in the entropic field profile) rather than merely accepting it as a given.

This identification connects to the broader program of the Theory of Entropicity, in which all physical constants and parameters are emergent from the entropic field. The speed of light c , Newton's constant G , Planck's constant $\hbar$, and Boltzmann's constant k_B are all, in the ToE framework, related to properties of the entropic substrate and the emergence map. The

identification of mass as entropic charge is one concrete instance of this broader program, and it provides the first quantitative bridge between the information-theoretic formalism and the measurable gravitational parameters of astrophysical bodies.

6.2. The Newtonian Limit of the Schwarzschild Metric

In the weak-field, Newtonian limit, the Schwarzschild metric yields the gravitational potential

(IE-S.102.15)

Φ

(

r

) =

–

GM

/

r

The derivation of the Newtonian potential from the Schwarzschild metric proceeds through the standard weak-field expansion. In the weak-field regime ($GM/(rc^2) \ll 1$), the metric component $A(r) = 1 - 2GM/(rc^2)$ can be written as

$$A(r) = 1 + 2\Phi(r)/c^2$$

where $\Phi(r) = -GM/r$ is identified as the Newtonian gravitational potential. This identification follows from the geodesic equation: in the non-relativistic limit ($v \ll c$), the geodesic equation for a test particle in the metric (IE-S.102.10) reduces to Newton's second law $d^2r/dt^2 = -d\Phi/dr$, with Φ defined by $g_{00} = -(1 + 2\Phi/c^2)$. The temporal metric component thus encodes the Newtonian potential, and the relation $A(r) = 1 + 2\Phi/c^2$ is the standard dictionary between the relativistic metric and the Newtonian potential.

The Newtonian potential $\Phi(r) = -GM/r$ is, of course, the solution of Poisson's equation $\nabla^2\Phi = 4\pi G\rho$ in the vacuum ($\rho = 0$ outside the source), with boundary conditions $\Phi \rightarrow 0$ as $r \rightarrow \infty$ and the total mass $M = \int \rho d^3x$ determined by the source. The Schwarzschild metric therefore contains the Newtonian potential as its weak-field limit — a consistency check that any relativistic theory of gravity must pass.

6.3. The Entropic Effective Potential and the Matching Procedure

On the other hand, in the ToE framework, the entropic field $S(r) = S_0 + S_1/r$ generates an effective potential via the emergence map and the GEFE/OFE. To leading order in the weak-field regime, we can write

(IE-S.102.16)

Φ

(

r

)

$\propto$ S 1 $/$ r

so that matching (IE-S.102.15) and (IE-S.102.16) gives

(IE-S.102.17)

 GM $\propto$ S 1

The matching procedure is performed in the *overlap regime* — the range of distances at which both the Newtonian approximation (weak gravitational field, $GM/(rc^2) \ll 1$) and the linear entropic approximation (small perturbation of the entropic field, $S_I/(rS_0) \ll 1$) are simultaneously valid. In this overlap regime, both descriptions are applicable, and the matching of the gravitational potential (IE-S.102.15) with the entropic effective potential (IE-S.102.16) yields the proportionality (IE-S.102.17).

The proportionality constant in (IE-S.102.17) depends on the normalization conventions of the entropic field and the Fisher-Rao metric. It absorbs the factors of C (the Fisher-Rao normalization constant from (IE-S.102.4)), λ (the

emergence constant), and the specific statistical model used to define g_i . The key physical result is that the mass M is *linear* in the entropic charge S_i — this linearity is a consequence of the linearity of the weak-field equations (both the Laplace equation for the Newtonian potential and the linearized MEE for the entropic perturbation) and is expected to hold in the weak-field regime. At strong fields (near the Schwarzschild radius), nonlinear corrections may modify this relationship, but the linear identification suffices for the present purposes.

The physical content of (IE-S.102.17) is striking: *the mass of a gravitating body is proportional to the "amount of entropy" concentrated in its entropic field profile*. More massive objects have stronger entropic fields — they concentrate more information, more microscopic degrees of freedom, more entropic charge. The gravitational pull of a body is not caused by some mysterious "gravitational charge" but by the informational content of its entropic configuration. This is the ToE realization of the old intuition, traced back to Jacobson (1995) and Verlinde (2011), that gravity may be an *entropic force* — a macroscopic effect arising from the statistical behavior of microscopic degrees of freedom.

6.4. The Mass-Entropy Identification

More explicitly, one can write

(IE-S.102.18)

M

$=$

$\propto$

S

1

where α is a constant determined by the coupling constants of the theory (including λ , ξ , and numerical factors from the Fisher-Rao normalization). The precise value of α depends on the detailed normalization of the entropic field and the statistical model used to define g_I , but the key point is that the Schwarzschild mass M is not a free parameter: it is an emergent quantity proportional to the entropic "charge" S_I of the underlying entropic field configuration.

The constant $\alpha = \alpha(\lambda, \xi, C)$ encodes the full chain of identifications from the entropic field to the physical mass. It involves:

- The emergence constant $\lambda = \hbar G/(4k_B c^3)$, which sets the scale of the information-to-geometry conversion.
- The Fisher-Rao normalization constant C ($= 1/2$ for the Gaussian model), which depends on the specific statistical model.
- The non-minimal coupling parameter ξ (set to zero in the present analysis, but entering at higher orders).
- Numerical factors arising from the matching of the entropic effective potential to the Newtonian potential.

In principle, α can be computed from first principles once the normalization of the entropic field is fixed — that is, once the precise relationship between $\sigma^2(r)$ and $S(r)$ is specified (not merely the proportionality but the proportionality constant) and the statistical model is chosen. This calculation, while important for quantitative predictions, does not affect the qualitative result: *mass is entropic charge*.

The result $M = \alpha S_I$ is structurally analogous to the identification of electric charge with the winding number of the entropic phase field, established in Letter IIA of the ToE Living Review Letters Series. In Letter IIA, the electric charge Q of a particle was shown to be proportional to a topological invariant (the winding number n) of the entropic phase field $\theta(x)$ around the particle. Both identifications — $M = \alpha S_I$ for mass and $Q = e \cdot n$ for electric charge — exemplify the ToE program of reducing all physical parameters to emergent entropic quantities. Mass and charge are not fundamental; they are different measures of the information content of the entropic field.

6.5. Connection to Black Hole Thermodynamics

The identification $M = \alpha S_I$ has a profound connection to one of the most remarkable results in theoretical physics: the Bekenstein-Hawking entropy formula. In 1973, Jacob Bekenstein proposed, on the basis of thought experiments involving black holes and the second law of thermodynamics, that a black hole should possess an entropy proportional to the area A of its event horizon:

$$S_{\text{BH}} = k_{\text{B}} c^3 A / (4 G \hbar)$$

In 1975, Stephen Hawking confirmed this formula by showing that black holes emit thermal radiation (Hawking radiation) at a temperature $T_H = \hbar c^3 / (8 \pi k_B G M)$, and the first law of black hole mechanics then yields the Bekenstein-Hawking entropy.

In the standard framework of GR and quantum field theory on curved spacetime, the Bekenstein-Hawking entropy is deeply puzzling. Why should the entropy of a black hole be proportional to the *area* of the horizon rather than the *volume* enclosed? What are the microscopic degrees of freedom that the entropy counts? These questions — central to the black hole information

paradox — have motivated decades of research in string theory, loop quantum gravity, and other approaches to quantum gravity.

In the ToE framework, the Bekenstein-Hawking entropy is not a mysterious coincidence but a *consequence* of the identification of mass as entropic charge. The area of the Schwarzschild event horizon is $A = 4\pi r_s^2 = 16\pi G^2 M^2 / c^4$. Using $M = \alpha S_I$, the area is proportional to S_I^2 , and the Bekenstein-Hawking entropy is proportional to S_I^2 — a quadratic function of the entropic charge. The entropy of the black hole is proportional to the *area* because the entropic charge S_I (which determines M) is related to the information content of the entropic field, which — in the holographic limit — scales with the area of the bounding surface rather than the enclosed volume. The Bekenstein-Hawking formula is thus a natural consequence of the entropic origin of mass, and the holographic principle (the area scaling of entropy) is built into the ToE framework through the Fisher-Rao construction.

7. Concluding Summary

7.1. The Schwarzschild Solution in the Theory of Entropicity

Summary Theorem: Emergence of the Schwarzschild Geometry from the GEFE/OFE

Thus, the Schwarzschild solution appears in ToE as:

1. **A static, spherically symmetric entropic configuration** $S(r) =$

$S_0 + S_I/r$ solving the MEE in the appropriate limit.

2. **An emergent spacetime metric** g_S obtained from the Fisher-Rao information metric via the emergence map.

3. **A classical limit of the GEFE/OFE** in which the vacuum equations reduce to the Einstein vacuum equations, yielding the Schwarzschild form for $A(r)$ and $B(r)$.

4. **A mass parameter** M identified in terms of the entropic field parameter S_I .

This worked example shows that, in the appropriate regime, the Theory of Entropicity reproduces the Schwarzschild solution of general relativity and interprets the mass of the black hole as an emergent entropic charge of the underlying information manifold.

7.2. The Chain of Emergence

The complete logical chain through which the Schwarzschild geometry emerges from the Theory of Entropicity can be summarized schematically as follows:

Stage	Construction	Result
1	Solve MEE for static, spherically symmetric $S(r)$	$S(r) = S_0 + S_I/r$ (Coulomb-like entropic profile)
2	Construct Fisher-Rao information metric from $S(r)$	$g_{I,rr} \propto [S'/S]^2$ (information geometry)
3	Apply emergence map $g_S = \lambda g_I$	$B(r) \propto g_{I,rr}$ (spatial metric from

Stage	Construction	Result
		information metric)
4	Solve vacuum Einstein equations (classical GEFE limit)	$A(r) = 1 - 2GM/(rc^2),$ $B(r) = [1 - 2GM/(rc^2)]^{-1}$
5	Match entropic charge to mass	$M = \alpha S_1$ (mass as emergent entropic charge)

This chain is the first concrete, fully worked example of the GEFE/OFE producing a known exact solution of Einstein's general relativity. It validates the entire machinery developed in Letter IE: the Minimal Entropic Equation, the Fisher-Rao information metric, the emergence map, the Curvature Transfer Theorem, and the identification of physical constants with entropic parameters. Each element of the ToE formalism contributes to the derivation, and the Schwarzschild solution emerges as a tightly constrained consequence of the framework — not as an assumption, not as an approximation, but as an *emergent classical limit* of the entropic dynamics.

7.3. Predictions Beyond the Classical Limit

Beyond the classical limit, the Theory of Entropicity predicts corrections to the Schwarzschild solution that distinguish it from standard general relativity. These corrections represent the genuine predictive frontier of the theory and provide targets for observational and experimental tests:

8. **Entropic dispersion near the horizon:** At distances comparable to the Schwarzschild radius, the entropic field enters the strong-field regime, and the LOA stress-energy (particularly the kinetic term $\partial_\mu S \partial_\nu S$) becomes non-negligible. This produces corrections to the Schwarzschild metric that can be parameterized in the post-Newtonian expansion and compared with observations.

9. **OCI-dependent dark curvature effects:** If the emergence map is not exactly conformal (e.g., if λ acquires a spatial dependence due to quantum corrections), the OCI tensor becomes non-zero and contributes additional "dark curvature" to the spacetime geometry. This dark curvature could manifest as deviations from the Schwarzschild metric at large distances, potentially relevant for galaxy rotation curves and the dark matter problem.
10. **Quantum corrections from the SOA:** The Seeley-DeWitt heat kernel expansion generates higher-curvature corrections (R^2 , $R_{\mu\nu}R^{\mu\nu}$, etc.) that modify the Schwarzschild geometry at distances comparable to the Planck length. These corrections are negligible for astrophysical black holes but could be significant for Planck-scale black holes and may play a role in the resolution of the curvature singularity at $r = 0$.
11. **Yukawa-screened corrections for non-zero entropic mass:** If the entropic mass m is non-zero, the entropic field profile is Yukawa-screened (equation (IE-S.102)), and the Schwarzschild metric acquires exponential corrections at distances $r \gg 1/m$. These corrections constitute a Yukawa modification of gravity that could be tested by solar-system experiments and gravitational-wave observations.

All of these corrections are *falsifiable*: they make quantitative predictions that can, in principle, be compared with observations. The Theory of Entropicity therefore goes beyond a mere reformulation of general relativity; it is an extension with genuine predictive content.

7.4. Future Directions

The Schwarzschild derivation presented in this appendix is the simplest application of the GEFE/OFE to a known exact solution. Natural extensions include:

- **The Kerr solution:** Deriving the rotating black hole metric from a stationary, axially symmetric entropic field configuration. This requires extending the Fisher-Rao construction to two-parameter statistical models and solving the MEE with axial symmetry.
- **The Reissner-Nordström solution:** Deriving the charged black hole metric by coupling the entropic field to the electromagnetic sector, using the results of Letter IIA.
- **The Kerr-Newman solution:** The general case of a rotating, charged black hole — the most general stationary black hole solution in GR (by the no-hair theorem).
- **Cosmological solutions:** Deriving the Friedmann-Lemaître-Robertson-Walker (FLRW) metric from a homogeneous, isotropic entropic field configuration. This connects the ToE to cosmology and provides an entropic interpretation of the cosmological constant, dark energy, and the expansion of the universe.
- **Gravitational wave solutions:** Deriving linearized gravitational wave solutions from perturbations of the entropic field, verifying that the GEFE predict the correct gravitational wave polarizations and propagation speed.

Each of these extensions represents a further validation of the GEFE/OFE and an opportunity to extract predictions that go beyond standard GR. The Schwarzschild derivation, as the first and simplest case, establishes the foundation upon which all subsequent derivations will be built.

8. Remarks

Remark K.1.1: Yukawa Corrections and Modified Gravity Phenomenology

The derivation can be generalized to non-zero entropic mass $m > 0$, yielding Yukawa-screened corrections to the Schwarzschild metric at distances $r \gg 1/m$. These corrections could be relevant for modified gravity phenomenology (MOND-like effects at galactic scales). Specifically, the entropic field profile $S(r) = S_0 + (S_1/r)e^{-mr}$ generates an effective potential $\Phi(r) \propto (1/r)e^{-mr}$, which transitions from Newtonian ($\Phi \propto 1/r$) at short distances ($r \ll 1/m$) to exponentially suppressed at large distances ($r \gg 1/m$). If the entropic mass scale m is comparable to the inverse of the galactic scale ($m \sim 1/r_{gal} \sim 10^{-23} \text{ eV}/c^2$), the Yukawa corrections could modify the rotation curves of galaxies, providing an alternative to cold dark matter. The quantitative analysis of this scenario requires solving the GEF with the Yukawa profile and computing the post-Newtonian parameters, which is deferred to future work.

Remark K.1.2: The Event Horizon as an Entropic Surface

The identification $M = \alpha S_1$ suggests that the Schwarzschild radius $r_s = 2GM/c^2 = 2\alpha GS_1/c^2$ is proportional to the entropic charge. The event horizon is therefore an *entropic surface* — a surface of constant entropic charge density. In the ToE framework, the event horizon is not merely a

causal boundary (the boundary of the region from which no signal can escape to infinity) but an *informational boundary*: it is the surface at which the entropic field perturbation S_I/r reaches a critical ratio with respect to the background S_0 . This informational interpretation of the event horizon resonates with the holographic principle and with the proposal (advanced by 't Hooft and Susskind) that the event horizon is a holographic screen encoding the information content of the interior.

Remark K.1.3: Fubini-Study Metric and Quantum Corrections

The derivation can be repeated using the Fubini-Study metric instead of the Fisher-Rao metric, yielding the quantum-corrected Schwarzschild metric. The Fubini-Study metric is the natural metric on the projective Hilbert space of quantum states, and it reduces to the Fisher-Rao metric in the classical limit (when the quantum state can be represented by a classical probability distribution). Using the Fubini-Study metric in the emergence map would produce a spacetime metric that incorporates quantum coherence effects — corrections to the Schwarzschild geometry arising from the quantum nature of the entropic substrate. These corrections are expected to be significant near the Planck scale ($r \sim l_P$) and could play a role in resolving the curvature singularity at $r = 0$. This connects the ToE to the program of quantum gravity and the information-theoretic properties of black holes.

Remark K.1.4: Uniqueness of the Entropic Configuration (Entropic

Birkhoff Theorem)

The uniqueness of the Schwarzschild solution (Birkhoff's theorem) translates in the ToE framework to a uniqueness theorem for the entropic field: the *unique* static, spherically symmetric solution of the MEE in the massless, minimal-coupling limit is $S(r) = S_0 + S_1/r$. This is the *entropic Birkhoff theorem*: among all static, spherically symmetric entropic configurations in the classical limit, only the Coulomb-like profile generates a consistent spacetime geometry through the emergence map. The proof is straightforward: the massless MEE in spherical symmetry reduces to the Laplace equation $\nabla^2 S = 0$, whose unique spherically symmetric solution regular at infinity is $S = S_0 + S_1/r$. The uniqueness of the entropic configuration and the uniqueness of the Schwarzschild geometry are thus two sides of the same coin — the former implies the latter via the emergence map, and the latter constrains the former via the GEFE.

Remark K.1.5: The Curvature Singularity and Its Entropic Resolution

The Kretschner scalar of the Schwarzschild metric, $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/(c^4r^6)$, diverges at $r = 0$. In the ToE framework, this curvature singularity corresponds to a divergence of the Fisher-Rao metric, suggesting that the entropic field $S(r)$ develops a topological singularity at the center. The resolution of this singularity may require the full quantum treatment (SOA + Vuli-Ndlela Integral), which regularizes the entropic field at short distances. Specifically, the SOA corrections

(higher-curvature terms in the GEFE) become dominant at $r \sim l_P$, modifying the Schwarzschild geometry and replacing the classical singularity with a regular core. The Vuli-Ndlela Integral — the non-perturbative completion of the entropic field theory — may further constrain the short-distance behavior by imposing topological conditions on the entropic field (e.g., requiring the information content to remain finite everywhere). The resolution of the Schwarzschild singularity within the ToE framework is a central open problem and a key test of the theory's self-consistency at the Planck scale.

9. Notation and Conventions

Symbol	Description
$S(r)$	Entropic field (scalar), function of radial coordinate
S_0	Asymptotic entropic background (value of S at spatial infinity)
S_1	Entropic charge (integration constant determining the field strength)
m	Entropic mass scale (from the potential $V(S) \approx \frac{1}{2}m^2 S^2$)
$\square$	Covariant d'Alembertian: $(1/\sqrt{-g})(\partial_\mu(\sqrt{-g} g^{\mu\nu} \partial_\nu))$
$V(S)$	Entropic potential
$\mathcal{G}_{l,rr}$	Radial component of the Fisher-Rao information metric
$\mathcal{G}_{S,\mu\nu}$	Physical spacetime metric

Symbol	Description
λ	Emergence constant: $l_P^2/(4k_B) = \hbar G/(4k_B c^3)$
Φ	Emergence map: $M_I \rightarrow M_S$
$A(r), B(r)$	Temporal and radial Schwarzschild metric functions
M	Schwarzschild mass parameter (emergent entropic charge: $M = \alpha S_1$)
r_s	Schwarzschild radius: $2GM/c^2$
ξ	Non-minimal coupling parameter
$R_{\mu\nu}$	Ricci tensor
$G_{\mu\nu}$	Einstein tensor: $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$
OCI	Obidi Curvature Invariant
LOA, SOA	Leading-Order Approximation, Second-Order Approximation
MEE	Minimal Entropic Equation
GEFE/OFE	Generalized Entropic Field Equations / Obidi Field Equations

10. Equation Index

Label	Equation	Section
(IE-S.100)	$\square S - V(S) = 0$ — Minimal Entropic Equation	§3 (Step 1)
(IE-S.101)	$(1/r^2) d/dr(r^2 dS/dr) - m^2 S = 0$ — Radial equation	§3 (Step 1)
(IE-S.102)	$S(r) = S_0 + (S_1/r) \exp(-mr)$ — Yukawa	§3 (Step 1)

Label	Equation	Section
	solution	
(IE-S.102.1)	$S(r) = S_0 + S_1/r$ — Coulomb limit	§3 (Step 1)
(IE-S.102.2)	$\sigma^2(r) \propto S(r)$ — Variance-entropy relation	§4 (Step 2)
(IE-S.102.3)	$g_I(\theta) = E[(\partial \ln p / \partial \theta)^2]$ — Fisher-Rao metric	§4 (Step 2)
(IE-S.102.4)	$g_{I,rr} = C[S'/S]^2$ — Radial information metric	§4 (Step 2)
(IE-S.102.5)	$S(r) = S_0 + S_1/r$ — Coulomb profile (restated)	§4 (Step 2)
(IE-S.102.6)	$S'(r) = -S_1/r^2$ — Derivative of entropic profile	§4 (Step 2)
(IE-S.102.7)	$g_{I,rr} \propto (S_1^2/r^4)/[S_0 + S_1/r]^2$ — Evaluated metric	§4 (Step 2)
(IE-S.102.8)	$ds_I^2 = g_{I,rr} dr^2 + r^2 d\Omega^2$ — Information manifold metric	§4 (Step 2)
(IE-S.102.9)	$g_{S,\mu\nu} = \lambda g_{I,\mu\nu}$ — Emergence map	§5 (Step 3)
(IE-S.102.10)	$ds_S^2 = -A c^2 dt^2 + B dr^2 + r^2 d\Omega^2$ — Schwarzschild ansatz	§5 (Step 3)
(IE-S.102.11)	$B(r) \propto g_{I,rr}(r)$ — Radial metric identification	§5 (Step 3)
(IE-S.102.12)	$R_{\mu\nu} - 1/2 R g_{\mu\nu} = 0$ — Vacuum Einstein equations	§5 (Step 3)
(IE-S.102.13)	$A(r) = 1 - 2GM/(rc^2)$ — Temporal metric function	§5 (Step 3)

Label	Equation	Section
(IE-S.102.14)	$B(r) = [1 - 2GM/(rc^2)]^{-1}$ — Radial metric function	§5 (Step 3)
(IE-S.102.15)	$\Phi(r) = -GM/r$ — Newtonian potential	§6 (Step 4)
(IE-S.102.16)	$\Phi(r) \propto S_1/r$ — Entropic effective potential	§6 (Step 4)
(IE-S.102.17)	$GM \propto S_1$ — Mass-charge proportionality	§6 (Step 4)
(IE-S.102.18)	$M = \alpha S_1$ — Mass-entropy identification	§6 (Step 4)

11. References

- [1] Schwarzschild, K. (1916). "Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie." *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 189-196. [English translation: "On the gravitational field of a mass point according to Einstein's theory."]
- [2] Birkhoff, G. D. (1923). *Relativity and Modern Physics*. Cambridge, MA: Harvard University Press.
- [3] Obidi, J. O. (2026). "Letter IE: On the Emergence of Physical Spacetime Geometry from Information Geometry." *Theory of Entropicity — Living Review Letters Series*.
- [4] Obidi, J. O. (2026). "Letter IIA: From 'The Question of $\mathcal{C}$ ' to the Rigorous Derivation of Maxwell's Electromagnetic Field Equations." *Theory of Entropicity — Living Review Letters Series*.
- [5] Obidi, J. O. (2026). "Letter IC: The Alemoh-Obidi Correspondence." *Theory of Entropicity — Living Review Letters Series*.

- [6] Jacobson, T. (1995). "Thermodynamics of Spacetime: The Einstein Equation of State." *Physical Review Letters*, **75**(7), 1260-1263.
- [7] Verlinde, E. (2011). "On the Origin of Gravity and the Laws of Newton." *Journal of High Energy Physics*, **2011**(4), 29.
- [8] Fisher, R. A. (1925). "Theory of Statistical Estimation." *Mathematical Proceedings of the Cambridge Philosophical Society*, **22**(5), 700-725.
- [9] Rao, C. R. (1945). "Information and the Accuracy Attainable in the Estimation of Statistical Parameters." *Bulletin of the Calcutta Mathematical Society*, **37**, 81-91.
- [10] Čencov, N. N. (1972). *Statistical Decision Rules and Optimal Inference*. Moscow: Nauka. [English translation: Translations of Mathematical Monographs, Vol. 53, American Mathematical Society, 1982.]
- [11] Bekenstein, J. D. (1973). "Black Holes and Entropy." *Physical Review D*, **7**(8), 2333-2346.
- [12] Hawking, S. W. (1975). "Particle Creation by Black Holes." *Communications in Mathematical Physics*, **43**(3), 199-220.

THE THEORY OF ENTROPICITY — LIVING REVIEW LETTERS SERIES

Supplementary Appendix K.1 to Letter IE

© 2026 John Onimisi Obidi. All rights reserved.

Document prepared May 2026.